Ruler CallTrack - Developer Brief (install & handover)

A self-hosted, multi-client **call-tracking platform** (CallRail alternative) for the Greek market. Node + SQLite + Twilio, with a branded web UI for managing clients and reading reports. Each client uses **their own Twilio credentials**, isolated per account.

1. What it does

- **DNI (dynamic number insertion):** swaps the phone number on a client's website per visitor, carrying the ad attribution (gclid/fbclid/utm).
- **Call tracking:** answered/missed, duration, first-time vs repeat caller, recording + transcription (with a Greek GDPR consent announcement).
- **Attribution -> conversions:** pushes qualified calls to **Google Ads** (offline click conversion via gclid), **Meta CAPI** (Lead), and **GA4**.
- **Branded admin UI:** add/manage clients (incl. their Twilio creds + conversion config + number pool) and view per-client reports (scorecards, channel breakdown, call log with recordings + transcripts).

2. Architecture

```
Browser snippet (dni-tracker.js) --POST /api/dni/assign--> Node app --> SQLite (data/cal
Twilio (per client) --POST /webhooks/voice/*--> Node app  (signature checked w/ that cli
Admin/staff --> /  (branded SPA)  --> /admin/api/*  (cookie-session auth)
                                   +-> conversion push: Google Ads / Meta CAPI / GA4
Caddy (reverse proxy) terminates HTTPS in front of the Node app.
```

- **Stack:** Node >=18, Express, better-sqlite3 (single-file DB), Twilio SDK, google-ads-api.
- **No external DB/service** required. One container does it all; Caddy adds auto-HTTPS.

3. Prerequisites

- A Linux server reachable on a public HTTPS domain (e.g. `calls.ruler.agency`). Free option: Oracle Cloud Always Free; cheap: Hetzner ~EUR 4/mo. (Full VM walkthrough in `DEPLOY.md`.)
- Docker + Docker Compose on that server.
- A subdomain DNS A-record pointing to the server.
- Per client: a Twilio account (or subaccount) with a Greek **regulatory bundle** approved and **Local (+30)** numbers purchased.

4. Install (Docker - recommended)

```
# on the server
git clone <your repo> ruler-calltrack   # or scp the phase2-call-tracking folder
cd ruler-calltrack
cp .env.example .env
nano .env            # fill the REQUIRED vars below
nano Caddyfile       # set your domain + email

docker compose up -d --build
docker compose logs -f app    # -> "ctc-call-tracking listening on :8080"
```

Open `https://<your-domain>/health` -> `{"ok":true}` with a valid certificate.

Required `.env` values

```
PUBLIC_BASE_URL=https://calls.ruler.agency   # exact public URL (used for Twilio signatu
BRAND_NAME=Ruler CallTrack                   # UI name
ADMIN_USER=admin
ADMIN_PASSWORD=<strong password>             # UI login
SESSION_SECRET=<openssl rand -base64 48>     # signs session cookies
# TWILIO_* are FALLBACK only - real creds are entered per-client in the UI
```

Run without Docker (alternative)

```
npm install
npm run migrate
ADMIN_PASSWORD=... SESSION_SECRET=... PUBLIC_BASE_URL=... npm start
# put Nginx/Caddy in front for HTTPS; keep alive with pm2 or a systemd unit
```

5. First-run configuration (via the UI)

1. Go to `https://<your-domain>/` -> sign in with `ADMIN_USER` / `ADMIN_PASSWORD`.
2. **New client** -> fill:

- **Basics:** client id (slug), name, forward-to number (their real line), lead value, qualified-after seconds, recording on/off, Greek consent text.
- **Twilio:** that client's **Account SID + Auth Token**.
- **Conversions (optional):** Google Ads `customerId`/`conversionActionId`, Meta `pixelId`/`capiToken`, GA4 `measurementId`/`apiSecret`.

3. Save -> add the client's **tracking numbers** to the DNI pool.
4. In **Twilio**, set each number's *Voice -> A Call Comes In* webhook to:

`https://<your-domain>/webhooks/voice/incoming` (HTTP **POST**).

5. Copy the **website snippet** shown in the editor onto the client's site.

> Secrets (Twilio token, CAPI token, GA4 secret) are write-only in the UI - they're stored in the DB and shown masked (`----1234`); leaving a secret field blank keeps the existing value.

6. Google Ads conversions (one-time agency setup)

Offline click-conversion upload needs agency-level OAuth. Set in `.env`: `GOOGLE_ADS_DEVELOPER_TOKEN`, `GOOGLE_ADS_CLIENT_ID`, `GOOGLE_ADS_CLIENT_SECRET`, `GOOGLE_ADS_REFRESH_TOKEN`, `GOOGLE_ADS_LOGIN_CUSTOMER_ID`. (Ask for `tools/google-oauth.js` to generate the refresh token.) Per-client `customerId`/`conversionActionId` go in the UI.

7. Verify the pipeline

```
# fake the whole flow without real telephony (set VALIDATE_TWILIO_SIGNATURE=false first)
docker compose exec app sh -c "VALIDATE_TWILIO_SIGNATURE=false BASE=http://localhost:808
# -> OK PASS - full attribution -> call -> qualify path works
```

Then place a real test call to a tracking number and confirm it appears in the UI report with duration, status, and a playable recording.

8. Security notes

- Admin UI is protected by a signed-cookie session (`SESSION_SECRET`); cookies are `HttpOnly`, `SameSite=Lax`, `Secure` under HTTPS.
- Every Twilio webhook is validated against **that client's** auth token (`X-Twilio-Signature`). Keep `VALIDATE_TWILIO_SIGNATURE=true` in production.
- Recordings are streamed through the app using the client's Twilio creds (browsers never get the token).
- Optionally set `REPORT_API_KEY` to require `X-Api-Key` on the read-only `/api/report/*` JSON endpoints (used for Looker/Metabase).

9. GDPR (Greece)

Recording requires informing callers. The app plays a Greek announcement before connecting (editable per client; default in `src/config.js`). Disable per client by unticking "Record calls". Transcripts are personal data - agree retention with each client. Have clients approve the wording.

10. Day-2 operations

- **Backup:** the DB is one file - `cp data/calls.db` off-box on a cron (it's the whole dataset).
- **Update:** pull/scp new code -> `docker compose up -d --build`.
- **Logs:** `docker compose logs -f app`.
- **Re-skin/rename:** edit the 6 brand variables at the top of `public/admin/styles.css` and `BRAND_NAME` in `.env`.

11. Cost

- Server + HTTPS: EUR 0 (Oracle Always Free + Caddy/Let's Encrypt) or ~EUR 4/mo (Hetzner).
- Twilio (per client): numbers ~EUR 1+/mo each + per-minute (inbound + forward leg). This is the only guaranteed cost; budget by call volume.